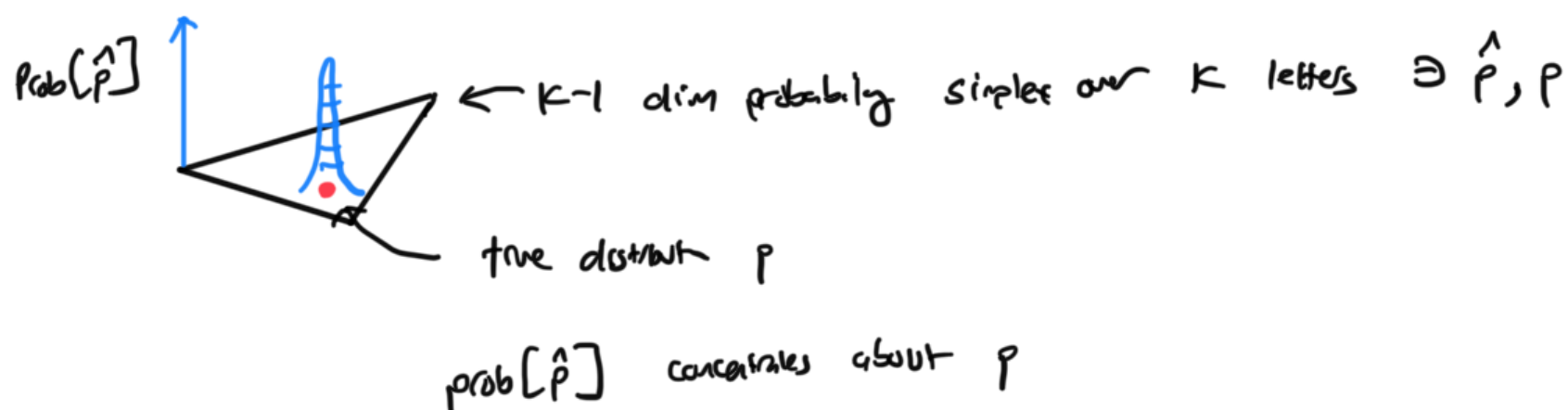


summary of important points from the last topic

- suppose you have a true distribution p over k letters $p_j \geq 0 \quad \sum_{j=1}^k p_j = 1$
- construct a long iid sequence $\vec{s}$ of length $N \quad \vec{s} = (s_1, s_2, \dots, s_N)$
- this sequence as an empirical distribution $\hat{p}_j(\vec{s}) = \frac{1}{N} \sum_{i=1}^N \delta_{s_i, j}$
 fraction of times letter j appears in sequence $\vec{s}$
- the empirical distribution $\hat{p}$ is a random distribution

$$\text{Prob}[\hat{p}] \approx e^{-N D_{KL}[\hat{p} \| p]} \quad \text{as } N \rightarrow \infty$$

$$D_{KL}[q \| p] = \sum_{j=1}^k q_j \ln \frac{q_j}{p_j}$$

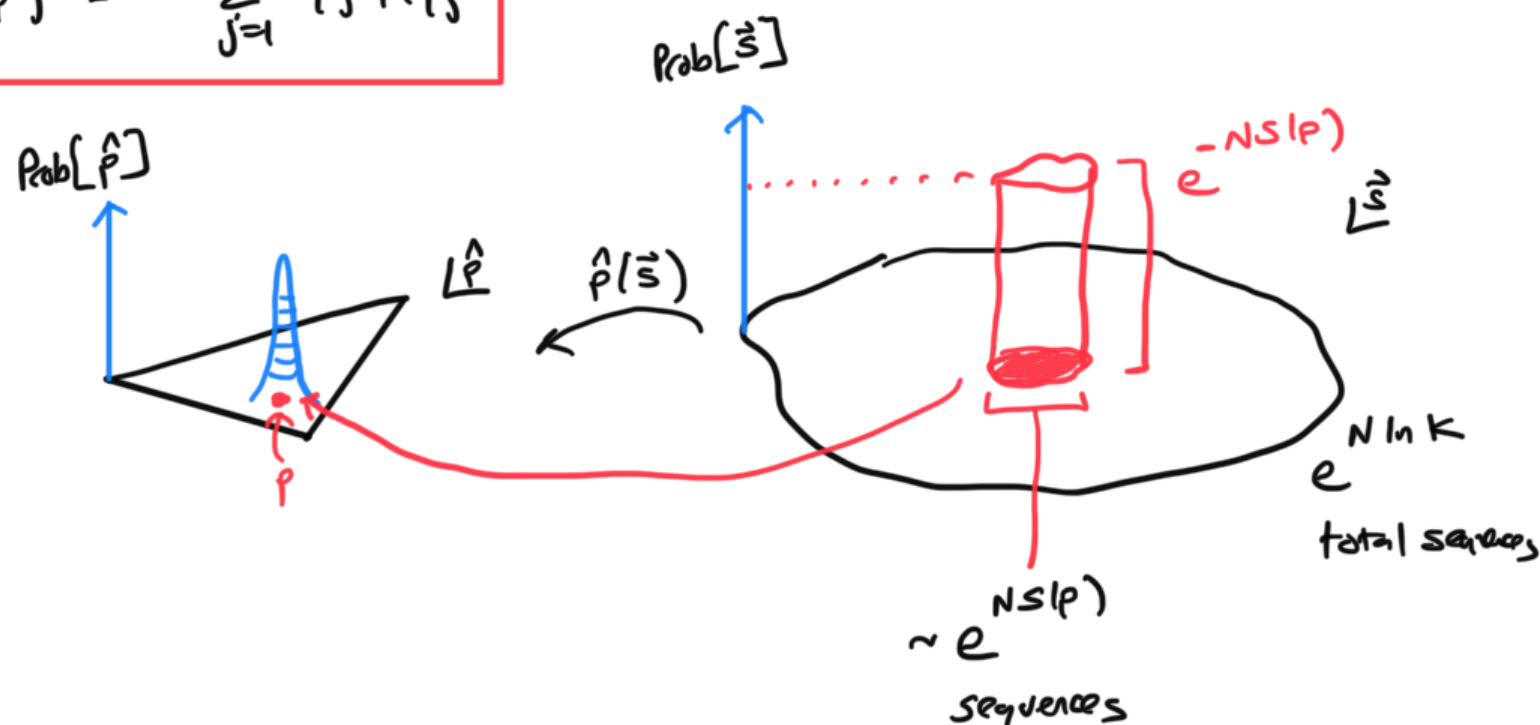


- Entropy How many sequences are there whose empirical histogram is p as $N \rightarrow \infty$

$$\Omega(p) \approx e^{N S(p)}$$

$$S(p) = - \sum_{j=1}^k p_j \ln p_j$$

• picture of $p(\vec{s})$



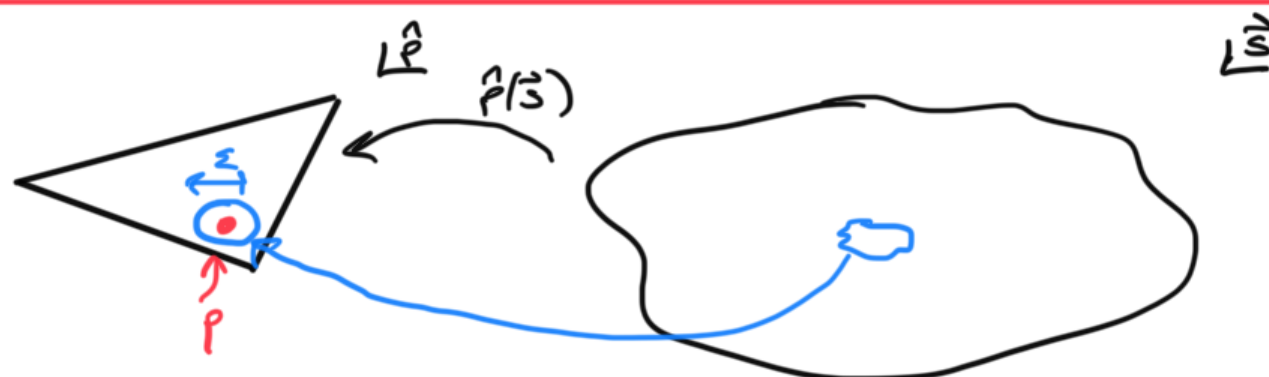
Asymptotic equipartition theorem: (AEP)

- as $N \rightarrow \infty$ almost all the probability mass of $p(\vec{s})$ is equally partitioned into $e^{NS(p)}$ typical sequences each with probability $e^{-NS(p)}$
- this is an exponentially small fraction of all $e^{N \ln k}$ sequences

more rigorous statement of AEP:

- intuitively a sequence $\vec{s}$ is typical if its empirical distribution $\hat{p}(\vec{s})$ is close to $p(\vec{s})$
- but how do we measure closeness? use the entropy S !

Def: Typical sequence $\vec{s}$ is ϵ -typical $\iff |S(\hat{p}(\vec{s})) - S(p)| < \epsilon$



let $T_{N,\epsilon}$ = set of ϵ -typical sequences of length N

Theorem (AEP)

- $\lim_{N \rightarrow \infty} \text{Prob}[T_{N,\epsilon}] = 1$
- for large enough N

$$e^{N[S(p) - \epsilon]} \leq |T_{N,\epsilon}| \leq e^{N[S(p) + \epsilon]}$$

typical sequences $\sim e^{NS(p)}$
- $\forall \vec{s} \in T_{N,\epsilon}$

$$e^{-N[H(p) + \epsilon]} \leq p(\vec{s}) \leq e^{-N[H(p) - \epsilon]}$$

all typical sequences have prob $\sim e^{-NS(p)}$

Also from last topic:

$$p(\vec{s}) = \frac{e^{-\beta E(\vec{s})}}{Z}$$

$$Z(\beta) = \int d\vec{s} e^{-\beta E(\vec{s})}$$

$$F(\beta) = -\frac{1}{\beta} \ln Z(\beta)$$

$$\beta = \frac{1}{kT}$$

Boltzmann
distribution

partition function

free energy

$$kT D_{\text{FL}}[q||p] = G[q] - F(\beta)$$

$$G[q] = \langle E \rangle_q - kT S(q)$$

$$\langle E \rangle_q \equiv \int d\vec{s} q(\vec{s}) E(\vec{s})$$

Gibbs free energy

Helmholtz

Free energy as a cumulant generating function

Introduction / review of cumulants

- suppose we have a distribution $p(s)$ $s \in \mathbb{R}$
- a way to understand $p(s)$ is through its moments:

$$\langle s^n \rangle_p \equiv \int ds p(s) s^n$$

- a nice way to encode the moments is through the moment generating function (MGF)

$$M_S(h) = \sum_{n=0}^{\infty} \langle s^n \rangle_p \frac{h^n}{n!} = \langle e^{hs} \rangle_p$$

$$\langle s^n \rangle_p = \left. \frac{d^n}{dh^n} M_S(h) \right|_{h=0}$$

n th moment $\leftarrow$ n th derivative of MGF

- now we often care about sums of iid random variables (ie to compute empirical averages)
- are moments additive under iid sums? NO !!

Let $S = S_1 + S_2$ S_1, S_2 drawn iid from $p(s)$

Is $\langle s^n \rangle_p = \langle s_1^n \rangle_p + \langle s_2^n \rangle_p$??

$n=1$ $\langle s \rangle_p = \langle s_1 \rangle_p + \langle s_2 \rangle_p$ $\checkmark\checkmark$ Linearity of expectation

$n=2$ $\langle s^2 \rangle_p = \langle (s_1 + s_2)^2 \rangle_p$
 $= \langle s_1^2 + 2s_1s_2 + s_2^2 \rangle$
 $= \langle s_1^2 \rangle + \langle s_2^2 \rangle + 2\langle s_1s_2 \rangle$
 $= \langle s_1^2 \rangle + \langle s_2^2 \rangle + \underbrace{2\langle s_1 \rangle \langle s_2 \rangle}_{\text{violates additivity}}$ xx !!

can we restore additivity by considering
a polynomial in the moments (ie cumulant)

$c_2 = \langle s^2 \rangle_p - \gamma \langle s \rangle_p^2$ choose γ to restore additivity

$c_2(s_1+s_2) = \langle (s_1 + s_2)^2 \rangle - \gamma \langle s_1 + s_2 \rangle^2$
 $= \langle s_1^2 \rangle + \langle s_2^2 \rangle + 2\langle s_1 \rangle \langle s_2 \rangle - \gamma [\langle s_1 \rangle + \langle s_2 \rangle]^2$
 $= \dots - \gamma \langle s_2^2 \rangle - 2\gamma \langle s_1 \rangle \langle s_2 \rangle$

$$\begin{aligned}
&= \langle s_1^2 \rangle + \langle s_2^2 \rangle + 2 \langle s_1 \rangle \langle s_2 \rangle - \delta \langle s_1 \rangle^2 - \delta \langle s_2 \rangle^2 \\
&= \underbrace{\langle s_1^2 \rangle - \delta \langle s_1 \rangle^2}_{c_2(s_1)} + \underbrace{\langle s_2^2 \rangle - \delta \langle s_2 \rangle^2}_{c_2(s_2)} + \underbrace{2(1-\delta) \langle s_1 \rangle \langle s_2 \rangle}_{\text{violation of nonadditivity of cumulants when } \delta=1}
\end{aligned}$$

so $c_2 \equiv \langle s^2 \rangle - \langle s \rangle^2$ Second cumulant !!

it is just the variance of s :

$$\text{Var}(s) = \langle (s - \langle s \rangle)^2 \rangle = \langle s^2 \rangle - \langle s \rangle^2$$

well known:
var is additive
under sums...

but how do we find all other cumulants?

• consider how the MGF transforms under sum

$$S = s_1 + s_2$$

$$\begin{aligned}
M_S(h) &= \langle e^{h(s_1+s_2)} \rangle \\
&= \langle e^{hs_1} e^{hs_2} \rangle \quad \leftarrow \text{property of exponentials} \\
&= \langle e^{hs_1} \rangle \langle e^{hs_2} \rangle \quad \leftarrow \text{independence} \\
&= M_{s_1}(h) M_{s_2}(h)
\end{aligned}$$

• MGF's multiply under addition $\Rightarrow$ moments not additive under addition

solution: take a log:

Def: cumulant generating function (CGF)

$$\phi_S(h) \equiv \ln \langle e^{hs} \rangle$$

by construction:

$$\phi_{s_1+s_2}(h) = \phi_{s_1}(h) + \phi_{s_2}(h)$$

so define cumulants:

$$c_n \equiv \left. \frac{d^n}{dh^n} \phi_S(h) \right|_{h=0}$$

by construction the cumulants are additive

c_n is a polynomial in moments of degree n

more generally : • let $\vec{s} \in \mathbb{R}^N$
 $p(\vec{s})$ be an arbitrary distribution for which MGF exists
 let $\theta(\vec{s}) \in \mathbb{R}$ be some observable

• $\Phi_\theta(h) \equiv \ln \langle e^{h\theta(\vec{s})} \rangle_p$ cumulant generating function for observable $\theta(\vec{s})$

calculate:

$$c_1 = \langle \theta \rangle_p$$

$$c_2 = \langle [\theta - \langle \theta \rangle]^2 \rangle$$

$$c_3 = \langle [\theta - \langle \theta \rangle]^3 \rangle$$

$$c_4 = \langle \theta^4 \rangle - 3 \langle \theta^2 \rangle^2 \quad \text{assuming } \langle \theta \rangle = 0$$

Example: gaussian

$$p(s) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(s-\mu)^2}{\sigma^2}} \sim \mathcal{N}(\mu, \sigma^2)$$

$$\Phi_\theta(h) = \mu h + \frac{1}{2} \sigma^2 h^2$$

$$c_1 = \mu$$

$$c_2 = \sigma^2$$

$$c_n = 0 \quad \forall n \geq 3$$

Application: quick proof of central limit theorem

• under scalar multiplication $s \rightarrow \gamma s$ $c_n \rightarrow \gamma^n c_n$

• let s_i for $i=1 \dots N$ be iid draws from a distribution $p(s)$ with all cumulants finite $c_1, c_2, \dots$

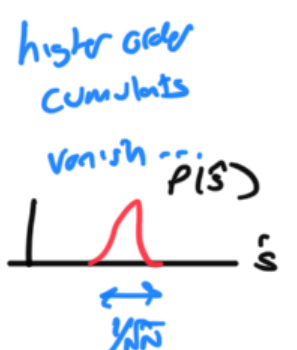
• consider sample average:

$$\hat{s} = \frac{1}{N} \sum_{i=1}^N s_i$$

• what are the cumulants of $\hat{s}$?

$$c_n(\hat{s}) = \frac{1}{N^n} N c_n$$

$$\Rightarrow \begin{aligned} c_1(\hat{s}) &= c_1 \\ c_2(\hat{s}) &= \frac{1}{N} c_2 \end{aligned}$$



• suppose now $c_1 = 0$ consider

$$\bar{s} = \frac{1}{\sqrt{N}} \sum_{i=1}^N s_i$$

$$\Rightarrow \frac{1}{N} N c_n \Rightarrow$$

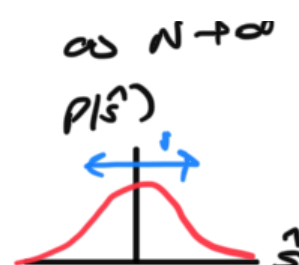
$$c_1(\bar{s}) = 0 \quad \rightarrow \quad \bar{s} \sim \mathcal{N}(0, 1)$$

$$c_n(\bar{s}) = \frac{1}{N^{n/2}} \dots$$

$$c_2(\bar{s}) = c_2$$

$$c_3(\bar{s}) = \frac{1}{\sqrt{N}} c_3$$

$$c_4(\bar{s}) = \frac{1}{N^{3/2}} c_4$$



aside What if cumulants don't exist?

people poor low down w/ heavy tails

wealth Musk 195B \$

Mike Bell 91B \$

Free energy and cumulant generating function

$$p(\vec{s}) = \frac{1}{Z(\beta)} e^{-\beta E(\vec{s})}$$

$$\Phi(\beta, h) \equiv \ln \langle e^{h\Theta} \rangle_p$$

$$= \ln \frac{1}{Z(\beta)} \int d\vec{s} e^{-\beta E(\vec{s}) + h\Theta(\vec{s})}$$

$$= \ln \frac{Z(\beta, h)}{Z(\beta, 0)}$$

$$Z(\beta, h) \equiv \int d\vec{s} e^{-\beta E(\vec{s}) + h\Theta(\vec{s})}$$

$$\Phi(\beta, h) = -[\beta F(\beta, h) - F(\beta, 0)]$$

$$-\beta F(\beta, h) \equiv \ln \int d\vec{s} e^{-\beta E(\vec{s}) + h\Theta(\vec{s})}$$

cumulant
generating
function

=

difference of free
energies!

to get cumulants we only care about derivatives w.r.t h

$\Rightarrow$

$-\beta F(\beta, h)$ also generates cumulants

physical intuition behind $F(\beta, h)$:

- take a physical system with energy $E(\vec{s})$
- turn on an external field h coupling the system to observable Θ

this changes the energy $E(\vec{s}) \rightarrow E(\vec{s}) - \underbrace{KT}_{\text{unit of energy}} \underbrace{h\Theta}_{\text{dimensionless}}$

- this changes the Boltzmann factor

$$-\beta E(\vec{s})$$

$$-\beta E(\vec{s}) + h\Theta$$

$$e \rightarrow e$$

- $F(\beta, h)$ is the free energy of the system with the field h turned on

Fluctuation response theorem

note: $\langle \theta \rangle = \frac{\partial}{\partial h} [-\beta F(\beta, h)] \Big|_{h=0}$

$\text{Var}(\theta) = \frac{\partial^2}{\partial h^2} [-\beta F(\beta, h)] \Big|_{h=0}$

$\Rightarrow \text{Var}(\theta) = \frac{\partial \langle \theta \rangle}{\partial h}$

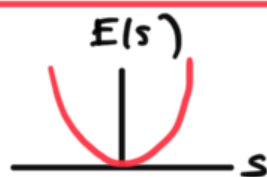
fluctuations when $h=0$

response when h is turned on...

Simplest possible example: charged particle in quadratic potential

$$s \in \mathbb{R}$$

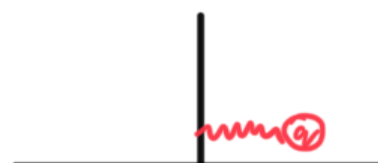
$$E(s) = \frac{1}{2} k s^2$$



particle connected to origin by a spring w/ constant k

k large = stiff spring

k small = loose spring



$$p(s) \propto e^{-\beta E(s)}$$

$$= e^{-\frac{\beta k s^2}{2}}$$

$$\langle s \rangle = 0$$

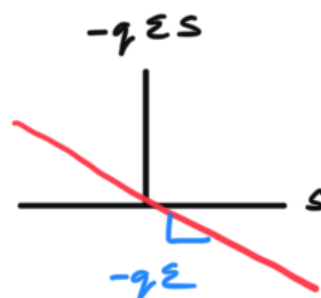
$$\text{Var}(s) = \frac{kT}{k}$$

now turn on an electric field:

$$E(s) \rightarrow E(s) - q\mathcal{E}s$$



+



q = charge

$\mathcal{E}$ = electric field

$h > 0 \rightarrow$
 $h < 0 \leftarrow$

$$-q\mathcal{E}s = -kT \left(\frac{q\mathcal{E}}{kT} \right) s$$

$\underbrace{\frac{q\mathcal{E}}{kT}}_h \underbrace{s}_\theta$
units inv dist dist

$$E(s) = \frac{1}{2} k s^2 - kT h s$$

$$\langle s \rangle = \arg \min E(s)$$

$$\frac{\partial E}{\partial s} = 0$$

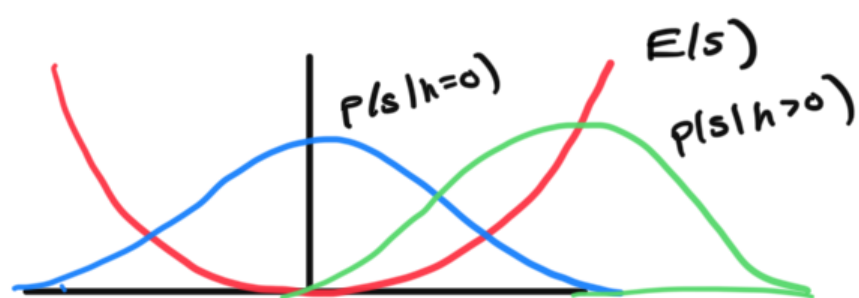
$$ks - kT h = 0$$

$$\langle s \rangle = \frac{kT}{k} h$$

$$\frac{\partial \langle s \rangle}{\partial h} = \frac{kT}{k} = \text{Var}(s) \therefore$$

Picture:

Small k : floppy / Soft spring



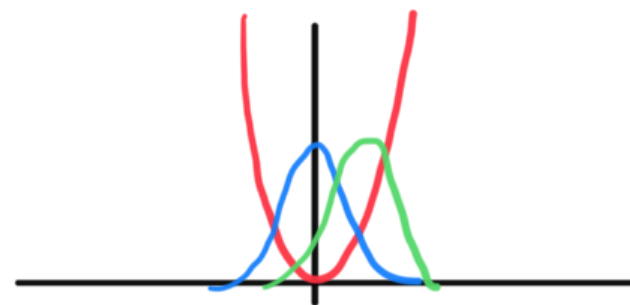
$$\text{Var}(s) = \frac{kT}{k} \text{ large}$$

Susceptibility to external electric field

$$\text{is also large} = \frac{\partial \langle s \rangle}{\partial h} = \frac{kT}{k}$$

$$\frac{\partial \langle s \rangle}{\partial \varepsilon} = \frac{\partial \langle s \rangle}{\partial h}$$

large k = stiff spring



$$\text{Var}(s) = \frac{kT}{k} \text{ small}$$

Susceptibility to electric field is also small

Final point: what about cumulants of the energy E

$$h \sigma \rightarrow h E$$

$$\phi(\beta, h) = \ln \frac{1}{Z(\beta)} \int d\vec{s} e^{\overbrace{-\beta E(\vec{s}) + h E(\vec{s})}^{-(\beta-h)E(\vec{s})}}$$

$$= \ln \frac{Z(\beta-h)}{Z(\beta)}$$

$$\phi(\beta, h) = -\beta [F(\beta-h) - F(\beta)]$$

$$C^n(E) = \left. \frac{\partial^n}{\partial h^n} \phi(\beta, h) \right|_{h=0} = \left. \frac{\partial^n}{\partial h^n} [-\beta F(\beta-h)] \right|_{h=0}$$

$$= (-1)^{n+1} \left[\frac{\partial^n}{\partial \beta^n} -\beta F(\beta) \right]$$

$$\boxed{\frac{\partial^n}{\partial \beta^n} [\beta F(\beta)] = (-1)^{n+1} C_n(E)}$$

$\beta F(\beta)$ generates (up to sign) cumulants of energy

Exercise: check via direct calc:

$$\underline{n=1} \quad \frac{\partial}{\partial \beta} (\beta F) = \langle E \rangle_p \quad \}$$

$$\boxed{\frac{\partial \langle E \rangle}{\partial \beta} = -\text{var}(E)}$$

$$\underline{n=2} \quad \frac{\partial^2}{\partial \beta^2} (\beta F) = -\text{Var}(E)$$

$$\frac{\partial^2}{\partial \beta^2} = -\text{Var}(E)$$

$\uparrow$
Response
 $\uparrow$
Fluctuations

inc $\beta \rightarrow$ lower $T \rightarrow$ dec $\langle E \rangle$

Identities involving Boltzmann distribution

$$p(\vec{s}) = \frac{1}{Z} e^{-\beta E(\vec{s})}$$

$$F(\beta) = -\frac{1}{\beta} \ln Z$$

$$\begin{aligned} S(\beta) &= - \int d\vec{s} p(\vec{s}) \ln p(\vec{s}) \\ &= - \int d\vec{s} p(\vec{s}) \ln \left[\frac{1}{Z} e^{-\beta E(\vec{s})} \right] \\ &= - \int d\vec{s} p(\vec{s}) [-\beta E(\vec{s}) - \ln Z] \\ &= \beta \langle E \rangle_p + \ln Z \end{aligned}$$

$$S(\beta) = \beta [\langle E \rangle_p - F(\beta)]$$

$$F(\beta) = \langle E \rangle_p - kT S(\beta)$$

Recall: $\langle E \rangle = \frac{\partial}{\partial \beta} (\beta F)$

